

Project Topic



Sentencing disparities
between the sexes

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Background Information – The Case of Aileen Wuornos

- ❑ Convicted and sentenced to death for killing Richard Mallory (Jan 31, 1992).
- ❑ 6 other trials followed; 5 also led to death sentences.
- ❑ Her case was highly publicized and issues subject to numerous documentaries
- ❑ Documentaries raises question on potential disparities and claims like women convicted of murder are sentenced heavier than men for similar crimes

“Are women sentenced more harshly than men for homicide cases?”

Importance

- ❑ Unequal sentencing impacts fairness, equality, and public trust.
- ❑ A truly impartial legal system must uphold justice regardless of gender.
- ❑ Significant differences in sentencing for similar crimes may reveal systemic bias.

Implications

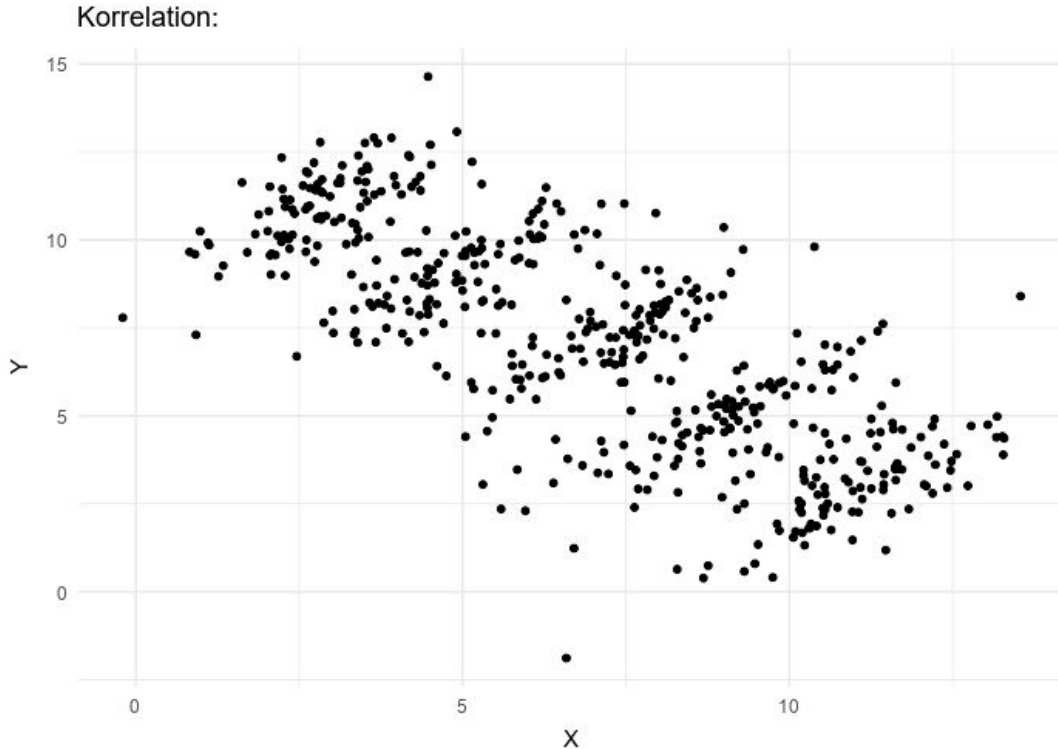
- ❑ Highlights potential gender bias in the criminal justice system
- ❑ Findings can support policy reform, judicial training, and public awareness.
- ❑ Ensures the justice system aligns with the core principle: equal justice for all.

Dataset

- ❑ **Dataset:** Criminal Justice Data Transparency (CJDT)
- ❑ FDLE Home / Criminal Justice Analytics Bureau
- ❑ Includes: Gender, race, age, type of conviction, sentence, defender type.
- ❑ Focus: Florida homicide cases.



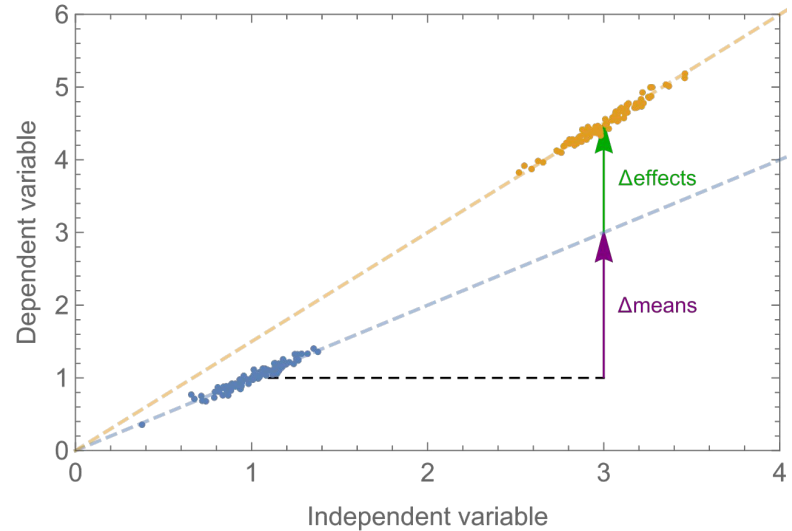
Statistical Caution: Simpson's Paradox



- ❑ Aggregate data can mislead (Homicide Cases)
- ❑ Must analyze **subcategories** (e.g., manslaughter vs. capital murder).
- ❑ Avoid drawing broad conclusions without controlling for variables.

Methods & Techniques : Oaxaca-Blinder Decomposition

- ❑ Originally used to determine wage differential between groups of male & female workers
- ❑ Breaks disparities into explained (e.g., case facts) and unexplained (bias) components.
- ❑ Structural modeling: Allows for scalability to other crimes like theft or assault.
- ❑ Control variables: Age, race, crime severity, legal representation.



Using Blinder-Oaxaca decomposition one can distinguish between "change of mean" contribution (purple) and "change of effect" contribution (green)

Broader Implications – Beyond Homicide

- ❑ While this project focuses on homicide cases, the ultimate aim is to develop a generalized, automated framework that can detect sentencing disparities across multiple types of crimes.
- ❑ The model will be built to scale across jurisdictions and adapt to different crime categories, such as:
 - ❑ Assault
 - ❑ Theft
 - ❑ Drug offenses
 - ❑ Burglary
 - ❑ Fraud